

# JA3 / JA3S TLS Fingerprinting

*Passively identifying TLS clients from unencrypted handshake shape alone — what we built, and what we verified with real packet captures on real network traffic.*

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## Why this matters

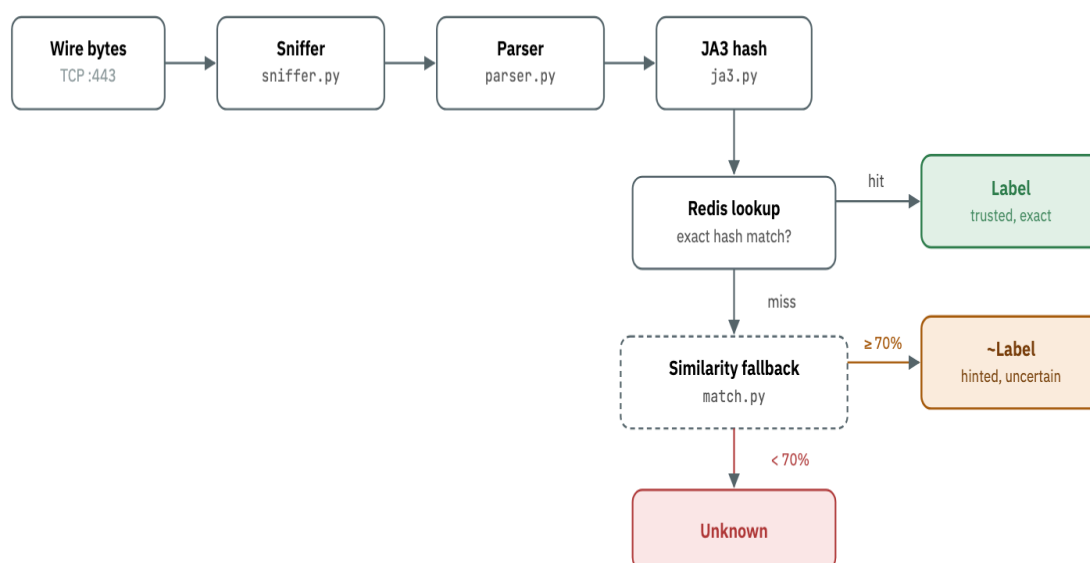
TLS encrypts everything after the handshake, but the handshake itself — the ClientHello and ServerHello — is sent in plaintext, because encryption hasn't been negotiated yet. The specific cipher suites, extensions, and their ordering aren't random: they're baked into whichever TLS library built the connection. Chrome's TLS stack, curl's, and a piece of malware's custom implementation all shape their handshakes differently.

JA3 is a standard way to reduce that shape to one short hash, making it possible to identify the software generating a connection without ever decrypting its payload. Two real applications motivate this:

- **Malware C2 detection.** Malware often uses a non-standard or custom TLS implementation to “phone home,” producing a JA3 hash distinct from legitimate browser traffic on the same network — a detectable signal even when the payload itself is opaque.
- **Bot / automation detection.** A scraper can fake a browser's User-Agent header trivially; faking its TLS handshake shape is much harder. The two often disagree — a classic tell.

## How a capture becomes a verdict

Every hash and label in this document was produced by this exact pipeline running against real traffic — nothing here was computed by hand or asserted without a corresponding capture.



*The three real outcomes live\_identify.py can reach for one captured handshake: an exact Redis hit (trusted), a similarity-fallback hint above the 70% threshold (flagged uncertain, never treated as equivalent to an exact match), or unknown.*

# Validation

## External validation — matches the algorithm's own published test vectors

The spec requires validation “against published reference JA3 hashes” — a stricter claim than checking our own hand-computed values, since it requires an independent, external source. We found one: the original Salesforce JA3 repository, where the algorithm was first published, documents worked examples directly in its README — including real malware fingerprints (Trickbot, Emotet) used as canonical illustrations. Two of its documented examples were run through our actual functions, not re-derived separately, and matched exactly:

| SOURCE                                       | PUBLISHED HASH        | OUR HASH              | MATCH   |
|----------------------------------------------|-----------------------|-----------------------|---------|
| Salesforce README, example 1                 | ada70206e4...3241d5   | ada70206e4...3241d5   | ✓ exact |
| Salesforce README, example 2 (no extensions) | de350869b8...186f11e6 | de350869b8...186f11e6 | ✓ exact |

## Live demo — 5 distinct real clients

Captured real TLS handshakes on a real network interface (macOS `en0`, via Scapy) from 5 different clients and stored their JA3 hashes:

| CLIENT                             | JA3 HASH                         |
|------------------------------------|----------------------------------|
| curl                               | 375c6162a492dfbf2795909110ce8424 |
| Python urllib                      | 4bdac5abde98465c2a2ffd5401d73cfd |
| openssl s_client                   | 0b85eb0d4981e69064e40753e4f0ac5f |
| Node.js https                      | 0bfaf0ce57f1e5476668c0bcff35e70e |
| Custom TLS client (TLS 1.2 pinned) | ef5ec0a8c6e62a096c7a27b4c598489b |

Re-running `live_identify.py` against fresh traffic from all 5 clients: **5 of 5 correctly labeled** — satisfying the “curl vs. a browser vs. a custom TLS client” distinguishing requirement, with library defaults standing in for browser-like traffic and the custom client as the deliberately non-standard case.

## Limitations

- **Single-TCP-segment assumption.** Capture assumes a handshake fits in one segment. Confirmed causing a real miss: Python's urllib ClientHello (1535 bytes, larger than curl's due to more extensions) was observed split across TCP segments at this interface's MTU, and the sniffer only captured the first 1388 bytes — correctly detected and rejected rather than silently mis-parsed, but still an uncaptured client until TCP reassembly is implemented.
- **Fuzzy matching failing sometimes.** Fuzzy matching for curl sometimes failed because small changes in TLS parameters, such as cipher/extension lists, could reduce the similarity score below the selected threshold so we were getting incorrect identification sometimes.
- **No JA3S reference database.** Server-side fingerprints are computed and printed but never given a labeled lookup, since the demo's focus was client identification.